KAVACH: System Architecture

A predictive NetJEPA and blockchain shield for India's critical networks · Team NetRunners

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1 Purpose, Scope and Quality Goals

KAVACH is an offline, world-model-based intrusion forecasting system. It ingests flow-level and packet-level traffic, learns how network state evolves over time, $P(S_{t+1} \mid S_t)$, forecasts attacker progression K steps ahead, maps behaviour to MITRE ATT&CK stages, explains every prediction, and records every decision on a tamper-evident ledger.

Users: SOC analysts, CISOs and CII operators (NCIIPC sectors). **Input:** PCAP / flow CSV. **Output:** risk timeline, stage, forecast, explanation, response plan.

Quality goals (priority order)

- **Predictive:** forecast attacks before escalation, not just classify them.
- **Explainable:** no black-box output; every alert carries evidence.
- **Sovereign & offline:** air-gapped, CPU-only, zero cloud calls.
- **Trustworthy:** tamper-evident decisions, models and training data.
- **Proportionate security:** controls scale with network classification.

2 High-Level Architecture (Building Blocks)

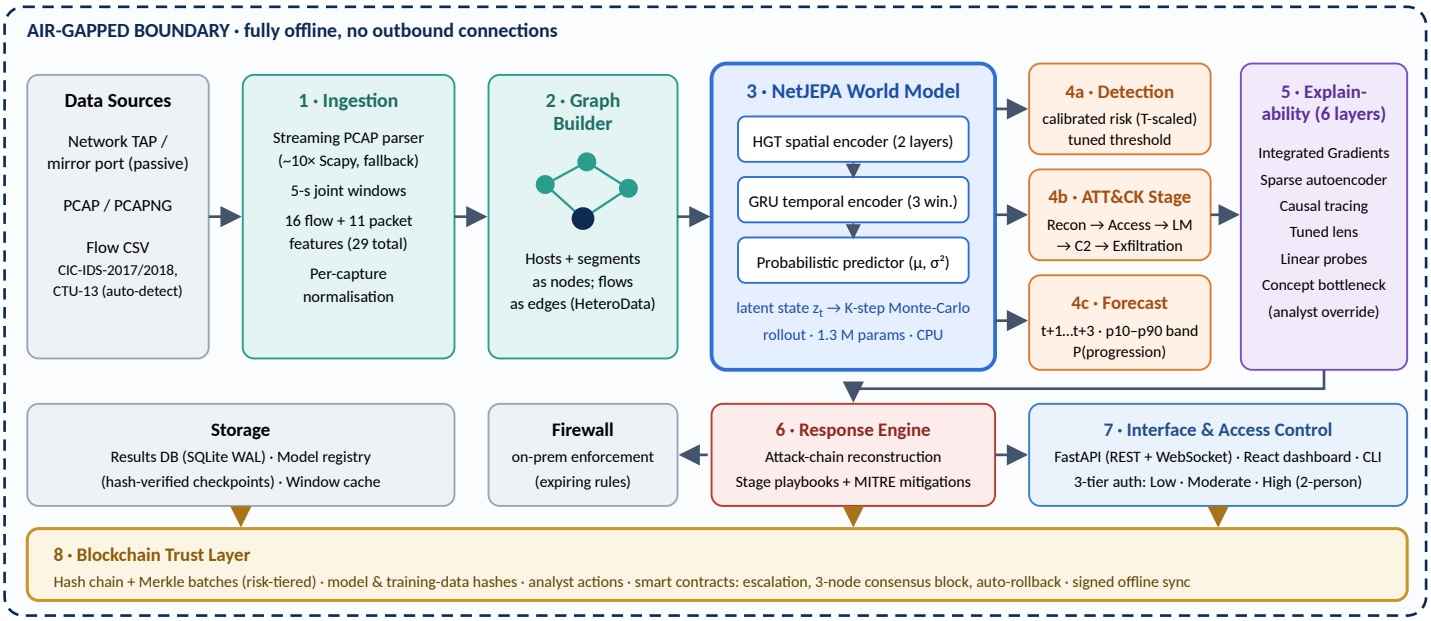


Figure 1. Building-block view. Data flows left to right; every stage writes evidence to the blockchain trust layer.

3 Runtime View: One Window, End to End

- **Parse (ingestion):** packets are decoded (IPs, ports, flags, TTL, window, fragments, payload, retransmissions) into 5-s windows; flow statistics use IPFIX-style timeouts so both levels describe the same slice.
- **Represent:** each window becomes a heterogeneous graph with a 29-value state vector; the previous two windows are held as context.
- **Encode & predict:** NetJEPA produces z_t and a Gaussian over z_{t+1}; 64 Monte-Carlo samples are rolled 3 steps ahead.
- **Score:** heads output calibrated infiltration risk, ATT&CK stage, the forecast band and progression probability for the current and each future state.
- **Explain & respond:** attributions and concepts are computed on the exact inference path; a prioritised playbook is generated.
- **Record & stream:** the result is chained to the ledger by risk tier, stored, and streamed live to the dashboard over WebSocket. The analyst's verdict is chained as well.

4 Key Architecture Decisions

Decision	Choice	Rationale
State representation	Heterogeneous graph per 5-s window	Keeps who-talks-to-whom structure flat vectors lose.
Dynamics model	JEPA-style latent prediction (HGT + GRU)	Learns $P(S_{t+1} \mid S_t)$ in latent space; trains on CPU.
Uncertainty	Gaussian predictor + Monte-Carlo rollout	Honest p10–p90 band (90% intervals cover 95%).
Supervision	VICReg + NLL + supervised transitions	Prevents collapse; forecasts as accurate as detection.
Trust	Local hash-chain ledger, Merkle batching	Air-gap native; Fabric-ready for multi-site.
Deployment	Passive TAP, CPU-only, on-prem	Not in traffic path; no GPU or cloud.

5 NetJEPA World Model: Training and Inference

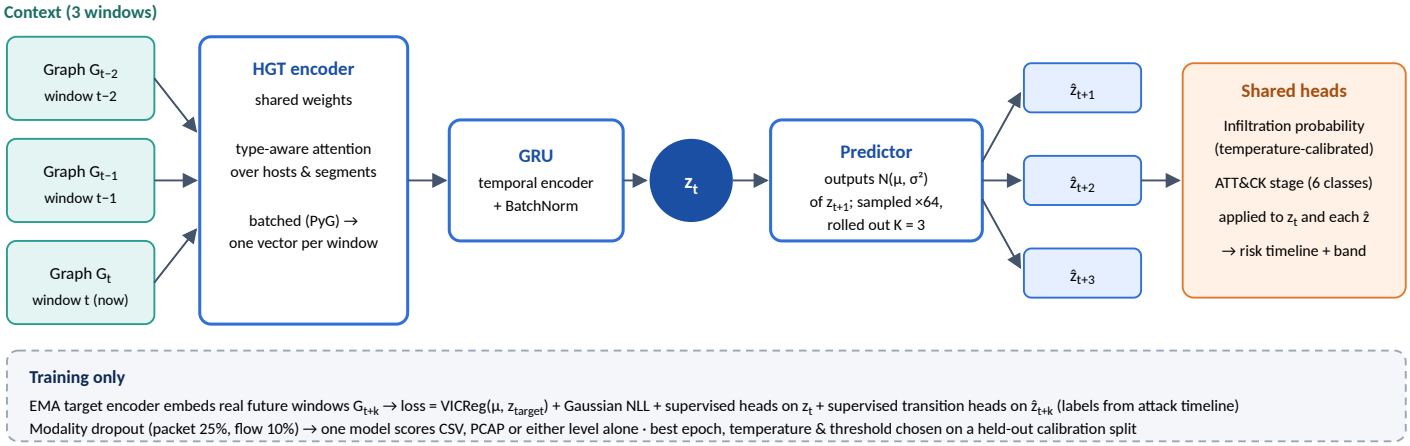
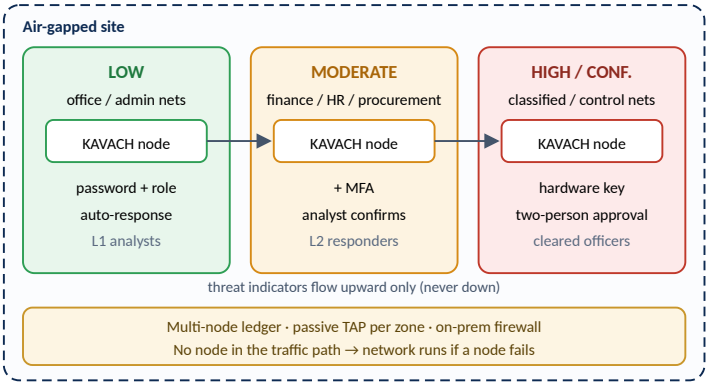


Figure 2. NetJEPA learns $P(S_{t+1} \mid S_t)$ in latent space and forecasts by rolling its own predictions forward.

6 Deployment View: Classification Tiers



7 Cross-Cutting Concerns

- **Security:** no outbound calls; header and statistics only (no payload); raw traffic never leaves the node.
- **Integrity:** every prediction, model load (weights hashed), training manifest and analyst action is hash-chained; any edit or deletion is detected on verification.
- **Model safety:** incompatible checkpoints are refused; training data is verified against its chained manifest; a model is auto-rolled back if its FPR on reviewed alerts exceeds 2× its predecessor's.
- **Feedback poisoning:** analyst verdicts are audited for suspicious volume, pattern and timing before retraining.
- **Compliance:** NCIIPC CII mandate (IT Act s.70A), CERT-In 6-hour reporting (explainable, timestamped records), DPDP Act 2023 data minimisation.
- **Reproducibility:** fixed seeds, config saved per checkpoint, 104 automated tests, one-command benchmark suite.

8 Technology Stack

Layer	Technology	Why
AI model	NetJEPA (PyTorch)	World model: predicts attacks, not only classifies; runs on CPU.
Graph learning	PyTorch Geometric (HGTConv)	Native heterogeneous graphs (hosts, segments, relations).
Ingestion	Custom struct PCAP parser, Scapy fallback	~10× faster than Scapy; streams multi-GB captures.
Explainability	IG, sparse autoencoder, tuned lens, probes	Explains the real inference path, not a surrogate.
Ledger	Hash chain + Merkle, Python smart contracts	Air-gap native; no internet or gas fees.
Backend / UI	FastAPI + WebSocket · React + TypeScript · CLI	Live streaming; dashboard with zero external requests.
Storage	SQLite (WAL)	Server-less and file-based, suited to air-gapped nodes.

9 Validation Summary and Known Limitations

0.86 vs 0.49	6.3% vs 29%	0.96–0.97	0.87 vs 0.62
F1 vs logistic-regression baseline	false-positive rate	forecast AUC, 1–3 windows ahead	F1, flow+packet vs flow only

Held-out, time-split validation on 8,043 windows of CIC-IDS-2017 (flows + raw PCAP). **Limitations:** leave-one-family-out tests show strong transfer to related families (unseen DDoS AUC 0.98) but weaker detection of dissimilar ones (PortScan AUC 0.58). Small feature perturbations can flip some detections. Lateral movement and exfiltration are rare in public data. **Mitigations:** rule and per-host baseline ensemble, adversarial training, and cross-dataset training on CSE-CIC-IDS-2018, CTU-13 and site traffic.